UESTC3031 - Engineering Project Management & Finance

(2024 – 2025 Semester 2)

Break-Even and Risk Analyses

(Assessment (Report))

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

Purpose of the Assignment

- *To demonstrate basics of engineering economics and break-even analysis / make-or-buy analysis*

Date Issued to Students: 05th May 2025 / Monday
Deadline / Submission Date: 15th May 2025 / Thursday
Submission: via Moodle

Submission Instructions:

- You must submit in **ONE pdf document** (not a zipped file, not two times, not in a word document) the **break-even analysis report** that accompanies it. You already have the instructions in the lecture notes and have been provided the **assignment answering file** that must be used.
- You must prepare this assessment in **pairs/groups of 2**. You must clearly indicate in the file name and in the document the student name's and IDs in the form **Student1_GUID_UESTCID_Student2_GUID_UESTCID.pdf** and inside the documents you will list a **table with the student's names, GUIDs, and UESTCIDs of your group members** so that we know who submitted with whom.
- You must submit a **plagiarism declaration** and your submission will go through **plagiarism check**.
- **No late submissions will be allowed in the Moodle** so if you don't submit you will get a zero for this complete assessment.
- This is a **SUMMATIVE assessment** (i.e. marked assessment).
- Only one student must submit the assessment prepared in groups so we must receive (via Moodle) only 1 submission in the system i.e. one for each group.
- While using references, you must use **IEEE style/format**.

Imran Shafique Ansari

Break-Even and Risk Analyses Report [100 Marks]

Ⅰ. Calculation [45]

You are a production engineer in an electronics company responsible for making electronic circuit boards. A customer has come to you with a design for a microcontroller and has requested you a quote for their business. Your manager has provided you with the documentation from the customer that includes a Bill of Materials (BOM) (as given in the Appendix). He has also provided you with a list of components (see Table I) required for producing the product and their costs from your approved suppliers (as given in the Appendix).

Your manager has requested that you provide a detailed cost analysis for a few different production volumes. *Examine your analysis based on the following:*

1. Pablo is one of the employees in your company, if he borrowed £5000 from a bank at 5% interest, three end-of-year payments will be made.

a. At the end of the first year, he will repay half of the £5000 principal plus the interest that is due.

b. At the end of the second year, he will repay half of the remaining £2,500 principal plus interest due.

c. At the end of the third year, he will repay the final £1,250 principal plus interest due.

Please write the relevant formula and calculate the amount he should repay each year and draw the corresponding Cash Flow Diagram. **[10/100]**

2. When producing a large number of PCBs (usually over 500), it takes a significant amount of time. To meet delivery deadlines, expedited shipping of raw materials is required to produce more than 500 PCBs.

Company A: The expedited shipping cost for raw materials per PCB is £1.5, with no additional charge.
Company B: The expedited shipping cost for raw materials per PCB is £1.3, but an additional £500 container rental fee is required.

Using the component cost figures provided, calculate the total cost to manufacture the following quantities of PCBs: 100, 1000, and 5000. Select the lowest cost method for each production quantity. Your answer must contain sufficient detail for your work/calculations to be checked by your manager. **[15/100]**

3. You have agreed a deal with your component suppliers, PCB manufacturers, and assembly operation on a total variable cost of £7.50 per board for all volumes between 0 and 5,000 boards. If your company has Fixed Costs of £19,500 and a selling price of £15.00, what is the Margin of Safety if you produce 4,000 boards? And what is the company profit when you produce 4000 boards? Draw a graph to illustrate this point. **[12/100]**

4. Your sales manager advises you that a new customer is willing to pay £27.00 per board but only for a maximum order of 1600 boards. How much profit or loss could you make on this order? Will there be a Margin of Safety in this case? If yes, calculate it. A graph is recommended. **[8/100]**

Hints:

1. We suggest you set up a **spreadsheet** to calculate the total costs for each production volume
2. Prices and price breaks are only for quantities of the same component.
3. Only products with identical parameters can be purchased in the same batch.
4. **Due to storage costs, excessive purchasing should be avoided**.

Ⅱ. Extensive Discussion [55]

5. Risk Management

Provide 8 possible risks involved in the project undertaken in the form of Risk Register and perform **quantitative analysis** for schedule and cost impacts of listed risks in the form of **influence diagrams** learned in the class, respectively. **[15/100]**

The following questions are based on the same background, for each (sub) question, the length must be no longer than 250 words.

6. Suppose you are the owner of Innovation Solution Inc., a start-up that has become highly successful by designing and selling specialized high-performance network routers **(Product Line A)** primarily to large enterprise data centers, the company is now planning to diversify its product range to create new profit points. The new product lines include **Product Line B**: Consumer-grade smart home hubs, which require faster development cycles, different marketing channels, and lower-cost manufacturing; and **Product Line C**: Custom IoT sensor arrays for specific industrial clients, which require close collaboration between specialized engineers, sales acting as consultants, and flexible operations for bespoke builds. How should the organizational structure of the company be adjusted to accommodate these new product lines and projects? **[10/100]**

7. For the custom IoT sensor arrays (**Product Line C**), what factors should Innovation Solution Inc. consider when deciding whether to make the sensors in-house or buy them from an external supplier? How would the company evaluate the need for control over the manufacturing process, quality, proprietary information, and the relationship with potential partners? **[10/100]**

8. Innovation Solution Inc. has diversified its product portfolio by adding consumer-grade smart home hubs (**Product Line B**) and custom IoT sensor arrays (**Product Line C**) to an already successful enterprise router business (**Product Line A**). Looking ahead:

(a) Identify and Analyze New Market Opportunities:
How should the company identify and evaluate emerging market trends, technologies, or customer segments to further expand its business? Consider the role of market research, competitive analysis, and customer feedback in determining the direction for future expansion.

(b) Role of Strategic Partnerships and Acquisitions:
Discuss how forming strategic partnerships, joint ventures, or pursuing mergers and acquisitions could facilitate rapid entry into new markets or product areas. What criteria should be used to select the right partners or targets, and how can these move support both short-term profitability and long-term innovation?

(c) Balancing Innovation with Scalability:
As Innovation Solution Inc. grows its product mix, what measures should be in place to ensure that increased diversification does not dilute its core competencies and innovation capabilities? How might investments in R&D or technology platforms contribute to a sustainable competitive advantage?

[10/100]

9. Given the current operational shifts—serving a mix of high-performance enterprise products, consumer electronics, and custom industrial IoT solutions—Innovation Solution Inc. will face challenges in balancing efficiency, flexibility, and control as it grows. Consider the following:

(a) Decentralization vs. Centralization:
Analyze the trade-offs between a centralized organizational structure and a decentralized (or matrix) organization for managing diverse product lines. How might each structure impact communication, decision-making, and responsiveness to market dynamics across the different business units?

(b) Scalable Organizational Design:
What organizational changes (such as establishing product-specific units, cross-functional teams, or dedicated innovation hubs) can support scalable growth as the company continues to expand? Evaluate the benefits and risks associated with each model in terms of maintaining quality, operational control, and proprietary advantages.

(c) Integration of Engineering and Finance Considerations:
In your answer, highlight how financial planning, budgeting, and performance metrics can be integrated within the organizational structure to align with strategic objectives. How can the company ensure that engineering innovations are efficiently transitioned to scalable production and market success without losing operational efficiency?

[10/100]

APPENDIX: Arduino UNO R3 BOM Costs

BOM Prices for 2025 (All prices in UKP)			
LM1117 (5.0V or 3.3V)			
Quantity			**Price (each)**
1+			0.916
50+			0.795
100+			0.605
250+			0.565
500+			0.525
1000+			0.364
2500+			0.356
5000+			0.349
Crystal (16MHz or 12MHz)			
Quantity			**Price (Each)**
1+			0.444
50+			0.327
100 +			0.269
250 +			0.260
500 +			0.251
1000 +			0.197
5000 +			0.192
ATMega 328 Microcontroller			
Quantity			**Price(each)**
1+			2.35
10+			2.06
25+			1.85
50+			1.72
100+			1.65
250+			1.49
500+			1.30
1000+			1.09
CH340G USB – Serial Interface			
Quantity			**Price(each)**
1+			0.507
10+			0.391
30+			0.367
100+			0.343
500+			0.332
1000+			0.320

USB Socket			
Quantity			**Price(each)**
5+			0.464
75+			0.324
150+			0.273
250+			0.216
500+			0.203
1500+			0.190
2500+			0.178
5000+			0.166

DC Power Socket			
Quantity			**Price(each)**
1+			1.441
10+			1.069
25+			0.983
100+			0.877
250+			0.856
500+			0.835

47uF / 25V Electrolytic SMD			
Quantity			**Price(each)**
10+			0.219
100+			0.149
500+			0.094
2500+			0.086
5000+			0.073
10000+			0.070
50000+			0.063

100nF / 10V ceramic 0603 SMD			
Quantity			**Price(each)**
10+			0.024
100+			0.013
500+			0.011
1000+			0.009
2000+			0.008
4000+			0.007
20000+			0.006

10K Resistor / 0603 SMD.			
Quantity			**Price(each)**
10+			0.010
100+			0.007
500+			0.004
1000+			0.003
2500+			0.002

LED SMD RED			
Quantity			**Price(each)**
5+			0.102
25+			0.054
100+			0.052
250+			0.045
500+			0.040
1000+			0.039

Single In Line connector (8 way) [Assume same cost for 6 way and 10way]			
Quantity			**Price(each)**
1+			0.722
10+			0.669
100+			0.508
500+			0.437
1000+			0.358
5000+			0.311
10000+			0.307

6-Way, Dual In-Line header, Through Hole			
Quantity			**Price (each)**
1+			0.480
10+			0.450
50+			0.415
500+			0.303
1500+			0.260
2500+			0.209

SMD Switch (Reset)			
Quantity			**Price**
10+			0.193
500+			0.166
1000+			0.132
5000+			0.124
10000+			0.119

PCB Cost: 2 layer, 53 x 68mm. (SEEED)			
Qty			**Price(UKP)**
5			6.15
10			9.40
50			44.38
100			69.04
500			247.70
1000			443.97
2500			990.05
5000			1805.48
Assembly costs: 28 components; 7 thru' hole, 18 SMT (PCBWAY)			
Qty			**Price (UKP) (for the given quantity)**
10			266.40
50			312.40
100			378.23
500			625.32
1000			785.38
2500			1600.02
5000			2374.05

Table I: *List of Components, and their Quantities*

Component	Description	Quantity
ATMega328	Microcontroller IC	1
CH340G	USB to Serial IC	1
LM1117-3.3	Linear Regulator IC (3.3V)	1
LM1117-5.0	Linear Regulator IC (5.0V)	1
Crystal, 12MHz- HC49 Package	USB Crystal	1
Crystal, 16MHz-HC49 Package	CPU Crystal	1
LED1 Red, SMT	LED, Power indication	1
LED2, Red, SMT	LED Indicator, Pin 13	1
C1, C2	Capacitor, electrolytic, 47uF, 25V	2
C3, C4, C5, C6	Capacitor, decoupling, 100nF, 10V, 0603	4
R1, R2, R3, R4, R7, R8	Resistor, 10K, 0603	6
SWT_1	Switch, NO, reset, SMT	1
CONN_1	DC Power Connector, Through hole	1
CONN_2	USB Connector, Through hole	1
SIP_1	8-Way Single In-Line Connector, Through Hole	2
SIP_2	6-Way, Single In-Line Connector, Through Hole	1
SIP_3	10-Way, Single In-Line Connector, Through Hole	1
SIP_4	6-Way, Dual In-Line header, Through Hole	1
PCB_1	Printed Circuit Board, FR-4, Double sided, 1.6mm (53mm x 68mm)	1
ASSY_1	Board Assembly costs; 28 Component, 7 Through Hole	1

---------- END OF ASSESSMENT ----------